In [13]:

```python
import pandas as pd
import numpy as np

def preprocess_data(input_path,output_path):
    df=pd.read_csv("Classes April-May 2018.csv")
    print("Original Shape:", df.shape)
    df=df.drop_duplicates()

    df.columns=(
        df.columns.str.strip().str.lower().str.replace("","_").
        str.replace("[^a-zA-Z0-9_]","",regex=True))
    for col in df.select_dtypes(include='object').columns:
        df[col]=df[col].astype(str).str.strip()

        df.replace("",np.nan,inplace=True)

        numeric_cols=df.select_dtypes(include=np.number).columns
        for col in numeric_cols:
            df[col]=df[col].fillna(df[col].median())
        categorical_cols=df.select_dtypes(include='object').columns
        for col in categorical_cols:
            if not df[col].mode().empty:
                df[col]=df[col].fillna(df[col].mode()[0])
            else:
                df[col]=df[col].fillna("Unknown")

        for col in df.columns:
            if "date" in col.lower():
                df[col]=pd.to_datetime(df[col],errors='coerce')

    for col in numeric_cols:
        Q1=df[col].quantile(0.25)
        Q3=df[col].quantile(0.75)
        IQR=Q3-Q1

        lower=Q1-1.5*IQR
        upper=Q3+1.5*IQR

        df=df[(df[col]>=lower)&(df[col]<=upper)]
        df=pd.get_dummies(df,drop_first=True)
        print("Cleaned Shape:", df.shape)

        df.to_csv(output_path,index=False)

        print("Preprocessing completed successfully")
        print("saved to:", output_path)

preprocess_data(
    input_path="Classes April-May 2018.csv",
    output_path="Classes April-May 2018.csv")
```

```
Original Shape: (2177, 7)
Cleaned Shape: (1816, 334)
Preprocessing completed successfully
saved to: Classes April-May 2018.csv
Cleaned Shape: (1816, 334)
Preprocessing completed successfully
saved to: Classes April-May 2018.csv
Cleaned Shape: (1790, 334)
Preprocessing completed successfully
saved to: Classes April-May 2018.csv
```

In [14]:

```python
import pandas as pd
import numpy as np

def preprocess_data(input_path,output_path):
    df=pd.read_csv("Classes June 2018.csv")
    print("Original Shape:", df.shape)
    df=df.drop_duplicates()

    df.columns=(
        df.columns.str.strip().str.lower().str.replace("","_").
        str.replace("[^a-zA-Z0-9_]","",regex=True))
    for col in df.select_dtypes(include='object').columns:
        df[col]=df[col].astype(str).str.strip()

        df.replace("",np.nan,inplace=True)

        numeric_cols=df.select_dtypes(include=np.number).columns
        for col in numeric_cols:
            df[col]=df[col].fillna(df[col].median())
        categorical_cols=df.select_dtypes(include='object').columns
        for col in categorical_cols:
            if not df[col].mode().empty:
                df[col]=df[col].fillna(df[col].mode()[0])
            else:
                df[col]=df[col].fillna("Unknown")

        for col in df.columns:
            if "date" in col.lower():
                df[col]=pd.to_datetime(df[col],errors='coerce')

    for col in numeric_cols:
        Q1=df[col].quantile(0.25)
        Q3=df[col].quantile(0.75)
        IQR=Q3-Q1

        lower=Q1-1.5*IQR
        upper=Q3+1.5*IQR

        df=df[(df[col]>=lower)&(df[col]<=upper)]
        df=pd.get_dummies(df,drop_first=True)
        print("Cleaned Shape:", df.shape)

        df.to_csv(output_path,index=False)

        print("Preprocessing completed successfully")
        print("saved to:", output_path)

preprocess_data(
    input_path="Classes June 2018.csv",
    output_path="Classes June 2018.csv")
```

```
Original Shape: (1112, 7)
Cleaned Shape: (923, 297)
Preprocessing completed successfully
saved to: Classes June 2018.csv
Cleaned Shape: (923, 297)
Preprocessing completed successfully
saved to: Classes June 2018.csv
Cleaned Shape: (907, 297)
Preprocessing completed successfully
saved to: Classes June 2018.csv
```

In [ ]: